# Measurement Reconstruction vs. Trajectory Reconstruction: Von Weizsäcker between Hermann and Popper

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Early discussions of quantum mechanics noted that the uncertainty relations prohibit exact trajectory predictions but seemed to leave room for retrospective trajectory reconstructions with arbitrarily sharp position and momentum. In his Chicago lectures, Heisenberg treated the physical reality of such reconstructed trajectories as a 'matter of taste.' In 1934, Popper seized on this concession and proposed a thought experiment that, by using conservation laws, sought to turn a sharp *trajectory reconstruction* into an exact trajectory prediction. Heisenberg and his assistants sharply criticized the proposal. This paper argues that in early 1935 Hermann, drawing on a structurally similar thought experiment, identified Popper's deeper confusion between context-independent trajectory reconstruction and context-dependent *measurement reconstruction*. It will be shown that the twenty-two-year-old von Weizsäcker, then Heisenberg's assistant, played a pivotal role both in criticizing Popper's arrangement and in providing the thought experiment on which Hermann relied. On the eve of the EPR debate, the Hermann–Popper controversy turned on whether one may legitimately combine counterfactual attributions of position and momentum into a single diachronic trajectory.

## Introduction

The status of particles' past trajectories became a significant point of contention in the early reception of quantum mechanics. In a letter to his students, Paul Ehrenfest reported an argument that Bohr had used in his discussions with Einstein at the 1927 Solvay conference (Ehrenfest to Goudsmit, Uhlenbeck, and Dieke, Nov. 3, 1927).[1] Bohr conceded that one may determine the position of a freely moving body at successive times and retrospectively calculate its intermediate velocity, and hence its momentum, with arbitrary accuracy. The uncertainty relations therefore appear to be violated. This, however, Bohr continued, is a misunderstanding. The momentum at the intermediate time has only been 'reconstructed,' not actually 'measured': "The uncertainty principle does not forbid calculating momentum from past data" (Ehrenfest to Goudsmit, Uhlenbeck, and Dieke, Nov. 3, 1927). As Bohr's (1928, 249) put it in the published Como lecture, the calculation of the particle's velocity between the two position measurements is a mere 'idealization' from which no meaningful, testable predictions can be derived.

[1] Cited in Kalckar 1985, 415–418, tr. in 37–41, 415–418.

On page 15 of the German edition of his Chicago lectures, Heisenberg (1930) noted in passing that the decision whether to believe in the physical reality of such past trajectories was, after all, a 'matter of taste' (*Geschmacksfrage*). Heisenberg probably did not anticipate that this cursory remark would become controversial. Schlick (1931) was the first to object on philosophical grounds, around the turn of 1931. If such a reconstruction of the past cannot in principle be subjected to possible verification through a testable prediction, then the reality of past trajectories is not a question to be answered as a 'matter of taste': the question itself is meaningless. A few weeks later, during a visit to Caltech, Einstein coauthored a short note in which he pressed the issue on physical grounds (Einstein, Tolman, and Podolsky 1931). In their two-particle thought experiment, Einstein and his coauthors (ETP) argued that an exact reconstruction of the past motion of one particle would make it possible, by appealing to the conservation of energy and the geometry of the setup, to predict in advance both the energy and the arrival time of the second particle. This would violate the uncertainty relations. They therefore concluded that any such exact reconstruction of the past trajectory must be forbidden.

As Ehrenfest pointed out to Bohr (Ehrenfest to Bohr, Jul. 9, 1931, AHQP/BSC 18), Einstein's thought experiment probably expressed a deeper concern[2]: by virtue of the uncertainty relations, the later experimental choice seemed to determine retroactively whether the distant emitted particle possessed a definite energy or a definite time of emission (see also Einstein 1932). Schrödinger's notes from this period, written after a talk by Einstein, record the same interpretation. Following a talk in Leiden in November 1931, Ehrenfest may have prompted Einstein to reformulate the argument in terms of position and momentum (Einstein to Ehrenfest, Apr. 5, 1932; Albert Einstein Archives, 10-231). According to Rosenfeld's recollection, Einstein presented a version of this argument at a talk in Brussels in 1933 (Rosenfeld 1967, 127–28). If Rosenfeld's recollection is trustworthy, Einstein had already developed something close to the EPR argument before his permanent departure for the United States on October of 1933.

By that time, the debate over quantum mechanics had begun to spread beyond quantum physicists and into the philosophical discussions of the German-speaking world. This broader reception was shaped by an obvious philosophical reference point: Kant's claim that causality is not an *a posteriori* contingent regularity discovered by experience, but an *a priori* necessary condition of the very possibility of scientific experience. It is therefore not surprising that the philosophical discussion of the new theory took the form of the 'causality debate' (Mehra and Rechenberg 2001, 2.VI.1). The question was not merely whether quantum mechanics required a revision of classical mechanics, but whether it had refuted the principle of causality itself (Frank 1932). The uncertainty relations challenge its validity by blocking epistemic access to a particle's future trajectory. The fact that they do not seem, however, to prohibit the reconstruction of past trajectories raised the further question of the particle's ontological status.

In 1933, Hermann (1933) was among the first to enter the debate over causality from the standpoint of the Kantian-Friesian philosophy outlined by Leonard Nelson (Soler 2017). Hermann initially sought to preserve the possibility of a deterministic completion of quantum mechanics. Her encounter with Heisenberg's group[3] and her

[2] See Bacciagaluppi and Crull 2024, Ch. 1 for more detail.

[3] Weizsäcker to Hermann, Dec. 17, 1933, Hermann to Weizsäcker, Jan. 14, 1934; Weizsäcker to Hermann, Jan. 30, 1934; Hermann to Heisenberg, Feb. 9, 1934; Herrmann 2019, Ab. III, Br.

stay at Leipzig in 1934 proved decisive in shifting her point of view. In particular, Heisenberg's assistant Carl Friedrich von Weizsäcker introduced her to his version of Heisenberg's (1927) $\gamma$-ray microscope. Weizsäcker's (1931) treatment suggested (Hermann to Bünning, May 23, 1934)[4] that in quantum mechanics retroactive inferences are licensed only within definite experimental arrangements: the electron's position can be inferred from the registration of the scattered photon only when the light is brought to an image plane, whereas a focal plane arrangement yields information about the electron's momentum. In this sense, Hermann concluded, quantum mechanics does not violate the principle of causality as such; it invalidates only its classical formulation in terms of prediction. Causality is preserved insofar as it allows for *measurement reconstruction*: from a registered mark on a scintillating screen, through the apparatus, back to the definite value of a quantity.

Popper entered the causality debate on quantum mechanics around the same time, probably in spring 1934, while completing *Logik der Forschung* (Giovanelli 2026). He aimed to interpret the indeterminacy relations as statistical dispersion relations and therefore denied that they set an intrinsic limit to the precision of measurements performed on an individual particle. After discussing the matter with Viennese physicists, especially Pauli's assistant Victor Weisskopf, Popper (1934a) devised a thought experiment also based on an electron–photon collision. In a manner not dissimilar to ETP (but with the opposite goal), Popper argues that if the electron's past path could be reconstructed from a momentum measurement followed by a position measurement, then conservation laws might be used to predict the future position-cum-momentum path of a single photon with arbitrary precision. Popper's argument thus relies on the possibility of *trajectory reconstructions*, which Heisenberg's 'matter of taste' remark seemed to leave open, and turns it into the basis for trajectory predictions. Popper aimed to show that the reality of particle trajectories cannot be denied, even if quantum mechanics provides only statistical laws for them. The principle of causality remains as a postulate: the requirement never to abandon the search for non-statistical laws describing those paths (Popper 1935, §78).

Popper and Hermann were born roughly a year apart and shared a similar Kant-Fries-Nelson philosophical background (Milkov 2012). They were both marginal figures, with little academic track record, although Hermann could rely on much more solid mathematical training. Both entered the causality debate around the same time, and they even met in person at the Eighth International Congress of Philosophy, held in Prague in 1934, while drafting their reflections on the new quantum mechanics. Yet they arrived at opposite conclusions. The opposition is especially sharp because their reflections centered on two thought experiments sharing certain structural similarities. Both Hermann and Popper focused on arrangements in which an electron and a photon collide, so that a quantity of one system can be inferred from a measurement performed on the other through conservation laws. Yet they put this common structure to divergent uses.

For Hermann, the point of the thought experiment is to show that quantum mechanics requires the *reconstruction of the measurement process*, but that this reconstruction is context-dependent: after an electron–photon collision, once the apparatus is configured to measure either the position or the momentum of the photon, one can determine either the position or the momentum of the electron. For Popper, by contrast, the point

457[3,4,6,7].

[4]Archive of the Friedrich Ebert Foundation.

of the thought experiment is to exploit the possibility of *reconstructing the particle trajectory* in order to make a trajectory prediction: after an electron–photon collision, if one can reconstruct both the past position and the past momentum of the electron with arbitrary precision, then, through the conservation laws, one can predict both the future position and momentum of a photon that the quantum formalism describes only statistically.

The twenty-two-year-old von Weizsäcker, as Heisenberg's assistant (Cassidy 2015), came to occupy a pivotal position in both interpretations of quantum mechanics. As the more philosophically inclined member of the Leipzig group, he became Hermann's main discussion partner, suggesting his version of the microscope thought experiment, which became her tool for analyzing the contextual conditions of measurement. On Popper's side, von Weizsäcker was the central figure in the Leipzig critical response to Popper's proposed thought experiment against the uncertainty relations in late 1934. It was von Weizsäcker who asked Hermann to review *Logik der Forschung*, which had just appeared at the turn of 1935. For this reason, in the final version of her, finished in March, Hermann included a criticism of Popper's interpretation of quantum mechanics that she also developed in her review (Hermann 1935b). Heisenberg and his assistants had dismantled, one after another, each version of the thought experiment that Popper submitted to them. Hermann tried to get to the heart of the matter. Popper relied on what Heisenberg had unwisely conceded: that the reality of particles' past trajectories might be treated as a 'matter of taste.' For Hermann, by contrast, the reconstruction of such trajectories was impossible as a 'matter of context.' *Trajectory reconstructions* are impossible because they combine inferences licensed by different and mutually exclusive *measurement reconstructions.*

With the exception of a few remarks in an otherwise broader-ranging paper by Frappier (2017), however, the Popper-Hermann debate has not been systematically examined nor von Weizsäcker's role appreciated. Popper and Hermann can be understood as proposing two ways of preserving *causality* after quantum mechanics: Popper through the reconstruction of the particle's path, Hermann through the reconstruction of the measurement context. Their thought experiments were presented shortly before Einstein and collaborators published the EPR thought experiment in May 1935 (Einstein, Podolsky, and Rosen 1935). Each anticipated its structure in a different way (Jammer 1974, 176–181). Both involved correlations between two systems, the use of conservation laws, and the possibility of inferring a quantity of one system by measuring another. With Einstein's 1935 paper, however, the center of gravity of the discussion shifted. The EPR discussion showed that the underlying issue was deeper and more general: the problem of trajectory reconstruction was a special case of the problem of state attribution. Hermann and Popper had discussed context-free 'trajectories' under the heading of 'causality'; in retrospect, however, this debate can be seen as a proxy for the Einstein–Bohr dispute over the 'reality' of context-free 'states.' In this new context, however, Popper's and Hermann's contributions had very different fates.

In correspondence, Einstein managed to convince Popper that his thought experiment was flawed (Einstein to Popper, Sep. 11, 1935; Albert Einstein Archives, 19-130). Popper did not return to quantum mechanics in a sustained way until the late 1950s (Del Santo 2019), when he began to suggest that the EPR setup could replace his original arrangement in demonstrating the possibility of reconstructing past trajectories with arbitrary precision. Hermann's thought experiment, by contrast, may be regarded as one of the earliest philosophical articulations of the line of response later developed in Bohr's and Heisenberg's reactions to EPR (Bacciagaluppi and Crull 2024, sec. 4.4). Popper's pronouncements on quantum mechanics, dismissed by the Leipzig group as the work of a physical *ignoramus*, became part of the oeuvre of a philosopher whose reputation grew after the war. Hermann's philosophical framework (Reichenberger 2026), by contrast, did not survive as a compelling general philosophy of science. Although taken very seriously in Leipzig, her penetrating and technically informed interpretation of quantum mechanics faded along with her philosophical framework, only to be rediscovered recently (Crull and Bacciagaluppi 2017; Crull 2022). The tables, however, now seem to have turned. Whereas Popper's attack on the Copenhagen interpretation has largely become a historical curiosity (Howard 2012), Hermann's interpretation of quantum mechanics has gained new visibility as one of its most sophisticated and coherent expressions (Bacciagaluppi 2025).

## 1 Popper's 1934 thought experiment Against the uncertainty relations

In November 1934 Arnold Berliner, the editor of *Die Naturwissenschaften*, sent Heisenberg the galley proofs of a short note on the uncertainty relations drafted by a then unknown Viennese philosopher. The note proposed a thought experiment meant to show that Heisenberg's inequalities

$$\Delta p_x \cdot \Delta x \geqq \frac{h}{4\pi}$$

could not simply be interpreted as limits on the accuracy of all possible measurements. Popper's starting point is the distinction between measurements usable for testable predictions and 'non-predictive' measurements (*nichtprognostische Messungen*). The usual thought experiments supporting the uncertainty relations, he claims, show only that certain more accurate measurements cannot be used for subsequent predictions. They do not show, without further argument, that every measurement of position and momentum is subject to the same restriction (Popper 1934a).

### *1.1 Popper's Thought Experiment*

Popper's proposed experimental arrangement (fig. 1) consists of two crossed beams. An electron beam $A$ and a light, X-ray, or $\gamma$-ray beam $B$ intersect at $S$. The beam $A$ is assumed to be a monochromatic parallel beam, and hence a plane wave. The beam $B$ passes through a narrow slit at $B_1$ and forms monochromatic spherical waves. Popper therefore treats both $A$ and $B$ as 'pure cases' (*reine Fälle*). He then considers a "mental" or "imagined" selection of two narrow partial beams $[A]$ and $[B]$ intersecting at $S$. These partial beams are not technically isolated from the larger beams by diaphragms or other devices, but are selected only in thought.

The momentum $\mathfrak{a}_1$ of the electrons in $A$ is known, as is the magnitude $|\mathfrak{b}_1|$ of the momentum of the light quanta in $B$. Hence, for the two 'mentally' selected partial beams $[A]$ and $[B]$, Popper assumes that the momentum vectors $\mathfrak{a}_1$ and $\mathfrak{b}_1$ before a possible collision at $S$ are known. One then considers those electrons in $[A]$ that, after the collision, travel in the direction $SX$. In the Compton effect, a high-energy light quantum transfers part of its momentum to an electron. If one knows the electron's momentum $\mathfrak{a}_2$ after the collision, the momentum vector $\mathfrak{b}_2$ of the light quantum in $[B]$ that collided with it at $S$ can be determined from momentum conservation: $\mathfrak{b}_2 = (\mathfrak{a}_1 + \mathfrak{b}_1) - \mathfrak{a}_2$.

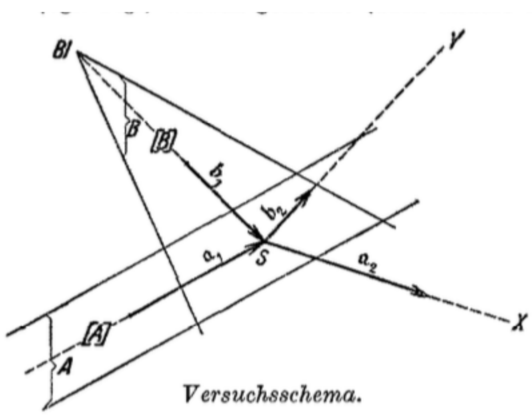


Figure 1: from Popper (1934a)

To determine $\mathfrak{a}_2$, Popper places a registering apparatus at $X$. In the 1934 note, he describes it as "an appropriately constructed point counter" (807). The term 'point counter' (*Spitzenzähler*) usually designates a Geiger-type counter that provides a relatively sharp practical determination of the place and time of a particle's passage. Popper, however, gives no technical details and assumes that an appropriately modified device could register both the magnitude of the momentum of the electrons arriving from the direction $SX$ and their time of arrival. For each registered value of $|\mathfrak{a}_2|$, Popper claims that one can reconstruct the electron's trajectory and determine the collision point $S$, which varies along the line $SX$ with $|\mathfrak{a}_2|$. Together with the registered arrival time, this reconstruction also determines the time of the collision. The position of the corresponding light quantum in $[B]$ at that time is therefore known, and its momentum $\mathfrak{b}_2$ after the collision can be calculated from momentum conservation. With this information, one can predict when the light quantum will arrive at $Y$, thereby predicting both its position and momentum with arbitrary precision.

The decisive premise is that the accuracy of non-predictive measurements, that is, the reconstruction of the past trajectory along $SX$, is not, in principle, subject to a restriction of the kind expressed by the uncertainty relations. Popper formulates this as a conditional assumption: he presupposes that the momentum magnitude and arrival time of an incoming particle can both be measured with arbitrary accuracy in a *non-predictive measurement*. He does not defend this premise in the note, but refers to a discussion elsewhere. The resulting determination of the position and momentum of the $B$-particle, however, has the character of a *predictive measurement*. One can place a second registering apparatus at $Y$, on the path of the $[B]$ particles scattered with the calculated momentum vector $\mathfrak{b}_2$, and test whether the predicted impacts occur at the calculated time. The experiment is thus meant to show that exact predictive measurements are possible, given the possibility of exact non-predictive measurements of position and momentum.

Popper emphasizes, however, that the arrangement does not allow one to assign an arbitrarily definite position and momentum to selected individual particles. The particles whose paths are predicted occur at randomly distributed intervals, and they cannot be technically isolated from the swarm of particles with randomly varying state quantities. The experiment therefore provides no means of producing a collection of particles more homogeneous than a 'pure case.' This qualification is central to Popper's intended conclusion. If Heisenberg's inequalities are interpreted statistically, as dispersion relations (*Streuungsrelationen*), then the experiment does not undermine them. It does not produce an ensemble of $B$ particles all having both sharp position and sharp momentum. If, however, the inequalities are interpreted as accuracy limits (*Ungenauigkeitsrelationen*) on measurements of individual particles, Popper claims that the thought experiment reveals their inadequacy. Exact predictions concerning individual particles appear possible, even though the statistical interpretation of quantum mechanics remains untouched.

### *1.2 Von Weizsäcker's Objection*

Heisenberg wrote to Popper on November 23, explaining that, at Berliner's request, he had discussed the note with his assistant von Weizsäcker. They then sent Berliner von Weizsäcker's brief rejoinder and asked him to forward it to Popper: "You will then see what kind of criticism it is. Essentially, it concerns the concept of the 'non-predictive measurement,' which in our view appears to be misunderstood in your work"(Heisenberg to Popper, Nov. 23, 1934; Karl Popper Archives 305-32). Heisenberg asked Popper whether he agreed with von Weizsäcker's considerations and expressed confidence that an agreement could be reached. Popper, who at the time had only one short publication to his name, was still unknown, and it is therefore quite remarkable that Heisenberg chose to engage with him, enlisting von Weizsäcker for the purpose—the most philosophically inclined of his collaborators.

First, von Weizsäcker dismisses the modified tip counter as a suitable device for measuring momentum at $X$ and suggests a scattering method instead. By measuring the Doppler frequency shift $\delta\nu$ of a light quantum scattered by the electron at $X$, one could infer the electron's momentum from energy and momentum conservation. The momentum measurement could then be followed by a tip counter and a clock mechanism recording the electron's position and arrival time. With this arrangement, von Weizsäcker concludes, the electron's 'trajectory' can be reconstructed only for the interval *between* the completion of the momentum measurement and the subsequent registration of position and time; it cannot be extended to the interval *before* the momentum measurement.

The difficulty is that measuring momentum by the Doppler effect requires determining the frequency of the scattered radiation with an uncertainty $\Delta\nu$ over a finite observation time $\Delta t$. A highly precise frequency determination requires observing many oscillation periods, making the time of the scattering interaction correspondingly uncertain. Greater precision in the momentum measurement therefore entails less precision in determining when that measurement occurred. Because the electron's velocity changes during the scattering process, its mean velocity over the duration of the momentum measurement is not exactly known. Consequently, even if its position after the momentum measurement is determined exactly, its position before that measurement cannot be inferred with arbitrary accuracy. The collision time at $S$ therefore remains uncertain, and the subsequent trajectory of the corresponding $B$-particle cannot be predicted with arbitrary precision. Von Weizsäcker concludes:

> This trajectory, however, is *in principle* uncontrollable. It is valid only for the time interval between the end of the momentum measurement and the beginning of the position measurement, during which the particle has no interaction whatsoever with its surroundings, and it cannot be extended into the period before the momentum measurement, since the latter itself destroys the knowledge of the position in accordance with the uncertainty relation. [...] Thus, even if the position [at $X_2$] after the momentum measurement is exactly known, one still cannot infer the position *before* the momentum measurement [at $X_1$] with an accuracy greater than the error $h/4\pi\Delta p$. That is, our

> knowledge of the trajectory of particle $A$ before the momentum measurement at $X$, and thus also of the trajectory of particle $B$ after the collision at $S$, is in accordance with the uncertainty relation. (Von Weizsäcker in, 808)

Von Weizsäcker concedes that this demonstration applies only to the *specific* experimental arrangement he has described. He nevertheless sees no reason to doubt that *any* other combination of measuring devices at $X$ would yield the same result. In his view, Popper fails to distinguish between the 'non-predictive measurements' he has defined and verifiable inferences about the past. Popper's 'non-predictive measurements' evade the uncertainty relations only because the statements reporting their results do not describe physically possible measurements. Verifiable inferences about the past, by contrast, remain subject to the same accuracy limits as verifiable predictions about the future, given the time-reversal symmetry of the laws of quantum mechanics.

## 2 The Popper-von Weizsäcker Correspondence

Popper replied on November 26, feeling that he had not been taken seriously. Drawing on the fuller analysis he had provided in Appendix VI of *Logik der Forschung*, which he attached to the letter (Popper to Heisenberg, Nov. 26, 1934; Karl Popper Archives 305-52), Popper argued that the past trajectory of a particle could be determined by three kinds of non-predictive measurement: (a) two position measurements; (b) a momentum measurement followed by a position measurement; or (c) a position measurement followed by a momentum measurement . Von Weizsäcker's considerations applied to cases (a) and (c): in these cases, the trajectory could indeed be reconstructed only for the interval between the measurements. Case (b), Popper insisted, was different:

> This line of reasoning, as far as I can see, is not entirely correct in one case: case (b), namely, a position measurement preceded by a momentum measurement. Admittedly, if the momentum measurement is performed here (cf. Mr. von Weizsäcker's note) by irradiating an individual particle with light, then it does indeed 'disturb' the position, and once again we can carry out the calculation only for the interval between the momentum measurement and the subsequent position measurement. The situation is different, however, if the momentum measurement consists in a momentum selection, for example, in the case of a light beam, by means of a filter, or, in the case of an electron beam, by means of an electric field perpendicular to the direction of the beam (an electron-velocity selector). In this case, we can not only calculate the exact 'trajectory' for the interval between the two measurements, but also trace it back into the 'past' (a claim discussed in greater detail in the printed enclosure). (Popper to Heisenberg, Nov. 26, 1934; Karl Popper Archives 305-52)

At $X$, a parallel-plate electrostatic analyzer producing a uniform electric field $E$ perpendicular to the beam direction (see fig. 2 below) filters a sub-ensemble of electrons with the same *predetermined* momentum $\mathfrak{a}_2$ *without changing it.* (Popper 1935, 220–21/299–300). If the momentum remains unchanged, then so does the position,[5] although the latter remains unknown. The unknown position may later be revealed by a second measurement at $X$, where a Geiger counter (or a clock-driven moving film strip) records the position and arrival time. Since the first momentum measurement leaves the electron's state unchanged, one might think that the electron's past trajectory could be reconstructed not only *between* the two measurements—momentum followed by position—but also *before* the first momentum measurement.

[5] Popper argues that otherwise this would imply action at a distance.

There was now too little time for Popper to prepare a rejoinder. On November 30, 1934, the Note appeared in volume 22 of *Die Naturwissenschaften*, together with von Weizsäcker's critical comment. On December 6, Popper sent Heisenberg a second communication (Popper 1934b) in response to von Weizsäcker, which he hoped to publish. Popper also attached a copy of the *Logik der Forschung* that had just been printed but not yet distributed. Popper's counterargument was based on a modification of the momentum measurement already proposed in Appendix VI of the *Logik der Forschung*. On December 6, the same day as Popper's letter, von Weizsäcker at Heisenberg's request, drafted a detailed response,(Von Weizsäcker to Popper, Dec. 6, 1934; Karl Popper Archives 360-21) which Heisenberg forwarded to Popper a few days later on December 10.

Von Weizsäcker's reply centered on Popper's proposed electrostatic analyzer (see fig. 2). Von Weizsäcker reconstructed it as a deflection experiment in which an electron beam passes between condenser plates $A$ and $B$ under the action of an electric field $E$. Without the field, the beam would follow the direction $DE$; with the field, it is deflected, so that each impact point on the screen $EF$ corresponds to a value of the transverse momentum. For small deflections, the displacement is

$$y = \frac{eE}{2m}t^2 = \frac{eE}{2m}\frac{l^2}{v^2}.$$

Popper's idea was that the impact point would provide a momentum measurement followed by a position measurement. Von Weizsäcker's objection was that the experiment's geometry already contains the limits expressed by the uncertainty principle.(Von Weizsäcker to Popper, Dec. 6, 1934; Karl Popper Archives 360-21)

To infer the momentum from the impact point, the electron's path through the field must be fixed with sufficient precision. This requires diaphragms at $D$ (fig. 2). But narrowing the diaphragm increases diffraction. If the slit has width $d$, then the transverse spread $\Delta y$ at the detector is constrained both by the slit width and by the transverse momentum uncertainty generated by diffraction:

$$\Delta y > d,$$

$$\Delta y = \frac{\Delta p_y}{mv}l,$$

with

$$d\Delta p_y \geq h.$$

It follows that

$$(\Delta y)^2 \geq \frac{h}{mv}l.$$

Thus the diaphragms required to define the beam path blur the screen point that is supposed to determine the momentum (Von Weizsäcker to Popper, Dec. 6, 1934; Karl Popper Archives 360-21)

This transverse uncertainty also affects the longitudinal motion. Since $y$ depends on the flight time $t = l/v$, an uncertainty in $y$ implies an uncertainty in $v$:

$$\frac{\Delta y}{y} = 2\frac{\Delta v}{v}.$$

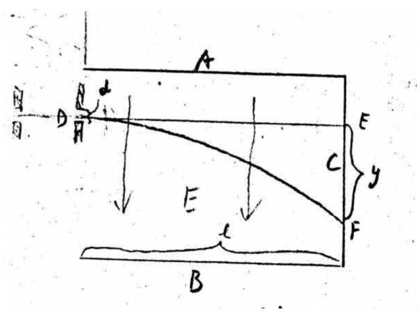

Figure 2: From the letter by von Weizsäcker to Popper, Dec. 6, 1934; KPA 360-21

This yields an uncertainty in the longitudinal momentum, $\Delta p_x = m\Delta v$, and in the reconstructed coordinate $x = vt$. Combining these uncertainties, von Weizsäcker concluded that the product $\Delta p_x \Delta x$ cannot be reduced below the order of Planck's constant. For small deflections, his calculation gives

$$\Delta p_x \Delta x > \frac{l^2}{y^2} h,$$

so that no violation of the uncertainty relation is obtained. Larger deflections require an additional angular correction, but do not alter the conclusion.(Von Weizsäcker to Popper, Dec. 6, 1934; Karl Popper Archives 360-21)

The decisive point was structural, not merely practical. The same arrangement that makes the impact point interpretable as a momentum measurement also prevents the earlier position from being sharply reconstructed. Von Weizsäcker therefore concluded that Popper had not supplied a realizable experiment capable of undercutting the uncertainty principle.(Von Weizsäcker to Popper, Dec. 6, 1934; Karl Popper Archives 360-21).

## 3 The Hermann-Weizsäcker Correspondence on Popper's Thought Experiment

In his reply of December 10, Heisenberg enclosed von Weizsäcker's criticism and then dismantled Popper's attempt to invert his thought experiment by replacing [$A$] with a light or Röntgen beam and using a light filter, rather than an electron spectrometer, to select light quanta that already possessed the 'right' momentum (Heisenberg to Popper, Dec. 10, 1934; Karl Popper Archives 305-32). Popper sent Heisenberg a revised version of the second communication summarizing the debate (Popper 1934c). He conceded that von Weizsäcker's argument against the electrostatic selector was correct but defended his idea of using a light-quantum filter against Heisenberg's objections (Popper to Heisenberg, Dec. 16, 1934; Karl Popper Archives 305-32). Heisenberg did not reply further. By then the Leipzig group had begun to tire of Popper's refusal to let the matter drop. But Popper persisted. He suggested another experimental arrangement, whose examination Heisenberg delegated to his other assistant Hans Euler. Popper envisaged two diffraction gratings inclined in opposite directions, each imposing a separate selection condition, hoping to select both wavelength and direction of the incoming light quanta. Euler replied on February 4, 1935 by showing that the two gratings form a single coherent interference system and cannot be treated separately.[6]

### *3.1 Hermann's Letter to Von Weizsäcker*

During those weeks, von Weizsäcker received a review copy of *Logik der Forschung* from the *Physikalische Zeitschrift*. Given their dispute in the preceding months, however, he declined to review the book and instead sent it to Hermann. Hermann wrote von Weizsäcker from Østrupgaard in Denmark to ask for clarification about his response to Popper's thought experiment (Hermann to Weizsäcker, Mar. 5, 1935; Herrmann 2019, Ab. III, Br. 14). In particular, she asked for an offprint of his reply to Popper in *Die Naturwissenschaften*, since she wanted to determine how far Popper had responded to

[6]Popper–Euler Correspondence, Karl Popper Archives 293-29.

his objections. She observes that Popper's thought experiment in the main text of *Logik der Forschung* appears essentially unchanged from the version he had explained to her in Prague, probably at the 1934 philosophy congress, and from the version discussed in *Die Naturwissenschaften.* The two appendices, however, seem to her to reflect Popper's discussion with Weizsäcker (Hermann to Weizsäcker, Mar. 5, 1935; Herrmann 2019, Ab. III, Br. 14). This was not in fact the case, since the Appendices $VI$ and $VIII$ had already been written before von Weizsäcker's response; nevertheless, as we have seen, Popper does seem to have anticipated a possible objection to his momentum-measuring apparatus and had already proposed a momentum filter as a replacement. Hermann, however, pointed out that this would not solve the problem:

> Both [appendices] concern the possibility of detecting a corpuscle in such a way that the position and time of impact, for example on a strip of film or by means of a point counter, are determined precisely, together with the direction and magnitude of the momentum with which the particle arrives. In the first case, the aim is to determine the magnitude of the momentum for a given direction; in the second, to determine the direction as well. What is more, the measurements are supposed to be such that they do not apply merely to the interval *between* the momentum measurement and the position measurement, during which, incidentally, no further disturbance may occur; rather, they are also supposed to determine the exact trajectory of the corpuscle for the time *before* the momentum measurement, so that, on the basis of some earlier collision, the trajectory of another particle can then be predicted with an accuracy exceeding the uncertainty relations. The decisive trick by which this physically inadmissible prediction is to be forced is, in the first appendix [VI], a momentum selection, that is, a selection of the particles in which this momentum is found, to be carried out on a nonmonochromatic, parallel-directed beam of particles, and specifically in such a way that the momenta of the selected particles, or their momentum component in the direction of the beam, are not changed. (Hermann to Weizsäcker, Mar. 5, 1935; Ab. III, Br. 14)

Hermann clearly identified the central flaw of Popper's arrangements: Popper needs to determine not only the electron's momentum between the momentum and position measurements, but also its momentum before the first momentum measurement, in order to reconstruct the collision time at $S$ and thereby predict the entire trajectory of the correlated particle $B$ testable at $Y$.

Hermann jotted down some still undeveloped ideas to ask for von Weizsäcker's opinion. They are articulated in three points.

1. Hermann first points out that, according to quantum mechanics, Popper cannot use the momentum measurement at $X$ to retrodict the earlier trajectory of the $A$-particle with arbitrary precision. At this point, she tentatively locates Popper's mistake in the assumption that a momentum component can be measured without an uncontrollable disturbance of that very component: "An exact momentum measurement will therefore be able to determine either the momentum before the measurement or the momentum after it, but not both" (Hermann to Weizsäcker, Mar. 5, 1935; Herrmann 2019, Ab. III, Br. 14). Otherwise, she argues, an exact momentum measurement would determine both the momentum before and the momentum after the measurement, thereby immediately violating the uncertainty relations. Yet Hermann does not rest content with this apparatus-specific disturbance argument. She asks whether the impossibility of such a measurement can be derived more generally from the deeper requirement of wave-particle duality. Popper appeals to various devices–a film strip or tip counter for registering the particle, a filter for a light beam, and spectral analysis in an

electric field for an electron beam–but Hermann is unsure how, in each case, the failure of the proposed apparatus follows from the necessary interplay of wave and particle descriptions.

2. Hermann then identifies the deeper source of Popper's mistake in his misunderstanding of wave-particle duality. From the *Logik der Forschung*, one can gather that Popper treats the $\psi$-function as a wave-like statistical description not of individual physical systems, but only of ensembles of such systems: "The error in this consideration is this: according to Heisenberg's derivation, the uncertainty relations follow from the dualism experiments and from the compulsion issuing from them to apply features of the wave picture to every atomic process, and features of the corpuscular picture to every wave propagation" (Hermann to Weizsäcker, Mar. 5, 1935; Ab. III, Br. 14). The uncertainty relations therefore limit not merely the composition of ensembles, but the applicability of classical state descriptions to individual systems. The probability interpretation of the wave function captures only one aspect of quantum mechanics: it specifies what can be said when one passes from one measurement context to another. After the momentum measurement, the system is in an eigenstate of the momentum operator; in this case the corresponding value is not merely probable but fixed. But if one asks what would be obtained by measuring position, for instance what position would be found after a momentum context has been fixed, the wave function supplies only a probability distribution over possible outcomes: "Popper fails to recognize this side of Bohr's complementarity" (Hermann to Weizsäcker, Mar. 5, 1935; Ab. III, Br. 14).

3. Given these premises, Hermann turns to past trajectory reconstructions. Popper is not merely unable to reconstruct the trajectory of the $A$-particle before the momentum measurement at $X$; it also makes no sense, in the relevant physical sense, to speak of a particle's trajectory between the momentum and the position measurement. In general, retrospective trajectory calculations are physically meaningless insofar as they combine determinations that belong to different observation contexts. These calculations ignore wave-particle duality by restricting themselves to the intuitive features of only one picture, usually the corpuscular picture, and then joining together quantities fixed in distinct measurements. The electron–photon interaction cannot be described solely as a corpuscular collision. The propagation of the scattered light must be described within the wave picture, while the interaction between the light quantum and the electron is described within the corpuscular picture. The apparatus is therefore needed not merely to register the collision, but to determine how the wave-optical propagation is correlated with the corpuscular scattering event. Optical instruments of this kind fix either the point of origin or the direction of propagation of the scattered light, but not both with arbitrary precision. Hermann here alludes to von Weizsäcker's $\gamma$-ray microscope. Hermann therefore concludes that Heisenberg's "much-debated remark", according to which the past trajectory was a matter of taste, was unfortunate.

As we shall see, von Weizsäcker would express some reservations about point (1). Otherwise, however, the letter contained the outline of Hermann's critique of Popper's interpretation of quantum mechanics, which she would articulate in both her long article on quantum mechanics (Hermann 1935a) and her review of *Logik der Forschung* (Hermann 1935b).

### *3.2 Von Weizsäcker's Reply to Hermann*

Hermann's tone is markedly different from Popper's self-assured manner. Despite having a much more sophisticated knowledge of quantum mechanics, she proceeds cautiously, asking at nearly every step whether von Weizsäcker would agree with her remarks. Von Weizsäcker thus plays the expert to whom Hermann turns for technical clarification (Weizsäcker to Hermann, Mar. 13, 1935; Herrmann 2019, Ab. III, Br. 15). For instance, at Hermann's request, von Weizsäcker explains what a tip counter is, an instrument with which Hermann was apparently unfamiliar. In such a counter, the sensitive electrode is a fine metal tip $S$ (*Spitze*), maintained at a high potential difference relative to the surrounding conductive wall $W$ of the gas-filled counter chamber. An ionizing particle passing sufficiently close to the tip produces primary ions in the gas; the strong electric field near the tip then amplifies this ionization into a discharge. The discharge registers that a particle has passed through the small sensitive region around the tip and thus gives a relatively sharp practical determination of the place and time of its passage. As von Weizsäcker adds, "A photographic plate, pinhole aperture, or the like therefore performs exactly the same function as the tip counter" (Weizsäcker to Hermann, Mar. 13, 1935; Herrmann 2019, Ab. III, Br. 15).

Figure 3: From Herrmann 2019, 485

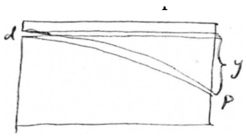

Figure 4: from Herrmann 2019, 483

The system, or any straightforward modification of it, is not suitable for measuring momentum, contrary to Popper's claim. For this reason, in his paper von Weizsäcker had proposed a measurement by the Doppler effect. He agrees with the result of Hermann's critique in point (1) of her letter, but slightly revises her diagnosis of the failure (Weizsäcker to Hermann, Mar. 13, 1935; Ab. III, Br. 15). On Hermann's diagnosis, an exact momentum measurement cannot determine both the incoming and the outgoing momentum, because the measurement must involve an uncontrollable disturbance of the measured momentum component itself (Weizsäcker to Hermann, Mar. 13, 1935; Ab. III, Br. 15). According to von Weizsäcker, however, even if the incoming and outgoing momenta are both known, an exact momentum measurement takes a finite time, so that the instant of the measuring collision is not known. Consequently, a later position cannot be traced back through the measuring process to an earlier position with arbitrary precision. Even granting Popper exact knowledge of the momentum before and after the measuring interaction, he still cannot reconstruct the collision time at $S$, because he does not know how long the particle moved with the pre-collision momentum and how long with the post-collision momentum (Weizsäcker to Hermann, Mar. 13, 1935; Ab. III, Br. 15).

Popper's selection of electron momenta in an electric field does not improve the situation. von Weizsäcker reiterates a simplified version of the argument that he and Heisenberg had developed in correspondence with Popper. He adds that Popper's filter has the same structure as the field-deflection measurement discussed by Heisenberg for a magnetic field in his textbook (Heisenberg 1930, 21–22). In Heisenberg's example, a beam of charged particles passes through a slit of width $d$, traverses a homogeneous magnetic field over a distance $a$, and is deflected by an angle $\alpha$, from which the longitudinal velocity, and hence the momentum, can be inferred:

$$\alpha = \frac{aHe}{mcv}.$$

Von Weizsäcker schematizes the same problem by considering a beam entering a field region through an aperture of width $d$ and arriving at a point $P$ after a displacement $y$. The arrival point can function as momentum information only because the field maps different momenta onto different deflections. But this inference presupposes that an aperture has defined the incoming beam: its finite width limits the geometrical accuracy of the reconstruction, while too narrow an aperture produces diffraction. Heisenberg therefore obtains, for the magnetic-field case, the constraints

$$\Delta\alpha \gtrsim \frac{d}{l+a} \qquad \Delta\alpha \gtrsim \frac{\lambda}{d},$$

which jointly prevent an arbitrarily sharp determination of both the beam localization and the direction of propagation. The same limitation applies to von Weizsäcker's simplified diagram: the later arrival at $P$ and the displacement $y$ do not allow one to reconstruct a precise earlier position independently of the conditions that made momentum selection possible. In both cases, the field-deflection measurement yields momentum information only at the cost imposed by aperture width and diffraction, so that the uncertainty relation is restored.

Popper had proposed light-filter arrangements as well, but von Weizsäcker notes that he initially had no clear account of what sort of filter should be used for light. Here again von Weizsäcker deploys the response that Heisenberg had already given to Popper. One possible realization would be resonance fluorescence: a gas transmits all light except for one spectral line, which, as Wood had shown experimentally, is almost metallically reflected. In this sense, the gas 'filters out' light quanta with a particular momentum. But this case exhibits the same difficulty as Popper's electron momentum selection. In resonance fluorescence, the atom is temporarily raised to an excited state with a finite lifetime. The sharpness of the spectral line is connected with the mean lifetime by the uncertainty relation, roughly:

$$h\Delta\nu \cdot \Delta t \sim h.$$

Thus the more sharply the spectral line, or equivalently the photon energy or momentum, is selected, the less sharply the time of the process is determined. The light quantum remains in the atom for a time uncertain by approximately $\Delta t$. This temporal indeterminacy plays the same role as the indeterminate moment of collision in the electron momentum measurement: it prevents one from combining an exact momentum determination with a precise temporal or positional reconstruction. Popper's filter therefore does not evade the uncertainty relations. Even when a filter seems to select a definite momentum component, the process by which it does so involves an uncontrollable temporal indeterminacy. This destroys the possibility of using the filtered photon in a precise trajectory-like reconstruction.

Von Weizsäcker recounts that, after the failure of Popper's first proposals, Popper devised further experimental arrangements. These, however, need not be discussed in detail. For a time, the refutation of Popperian arrangements had become, as von Weizsäcker put it, "a parlor game [*Gesellschaftsspiel*] for us in Leipzig, to refute Popper's setups" (Von Weizsäcker to Hermann, Mar. 13, 1935; Herrmann 2019, Ab. III, Br. 15).

The basic idea behind these refutations, he suggests, is this: the corpuscular picture is applicable only as long as one can also replace 'corpuscle' by 'wave packet.' Any arrangement capable of undercutting the uncertainty relation would have to make possible a wave group with

$$\frac{\Delta x}{\Delta \lambda} < 1.$$

Hence, if such an experiment is interpreted purely wave-theoretically, one may be sure of finding in it a contradiction with intuitive wave kinematics. For von Weizsäcker, moreover, an 'impulse selection' is not different in principle from an 'impulse measurement,' because of the symmetry of nature with respect to past and future. Yet he remains cautious about whether this "recipe for refuting Mr. Popper" can be transformed into a compelling general argument. In the end, he suggests, one must appeal to the quantum-mechanical formalism as a whole, which Popper seems not to have understood (Weizsäcker to Hermann, Mar. 13, 1935; Ab. III, Br. 15).

With respect to point 2, von Weizsäcker says that he can fully agree with Hermann's presentation, although his lack of familiarity with Popper's book prevents him from making the claim with complete confidence: "With Popper, it has already happened to me once that, with the best will in the world, I could not quite work out *what* he had actually misunderstood" (Weizsäcker to Hermann, Mar. 13, 1935; Herrmann 2019, Ab. III, Br. 15). At that time, Popper had assumed that his 'non-prognostic measurements' were "a generally known and accepted matter" (Weizsäcker to Hermann, Mar. 13, 1935; Herrmann 2019, Ab. III, Br. 15). Hermann's question was whether Popper's mistake lay in treating the wave function statistically, as if wave–particle duality applied only to ensembles rather than to individual quantum processes. Here, however, von Weizsäcker cannot satisfy Hermann's request to provide an experiment in which wave-particle duality is manifested in a single system. In processes involving individual particles, wave–particle dualism can hardly be demonstrated except by repeating the experiment. The reason may be that all measurements ultimately culminate in the determination of a position at a time, that is, a spacetime coincidence, thereby allowing a corpuscular description of the individual case.

## 4 Hermann on Popper in *Die naturphilosophischen Grundlagen der Quantenmechanik*

On March 19, Heisenberg wrote his last letter to Popper, explaining that all the considerations developed in his previous letter applied to transmitted light as well (Heisenberg to Popper, Mar. 19, 1935; Karl Popper Archives 305–32). With this, the Leipzig "parlor game" had run its course. Yet one might say that the official public answer of the Leipzig group to Popper came not from Heisenberg or von Weizsäcker, but from Hermann. In his reply to Hermann, von Weizsäcker does not comment on point (3) of her letter, although this point would become the core of her published refutation. Hermann's crucial claim was that quantum mechanics allows the reconstruction of measurement processes, but that such reconstructions must not be confused with the reconstruction of real trajectories. The formalism permits one to infer, under specified experimental conditions, what can be said about a past measurement event; it does not thereby license the attribution of a determinate path to the particle independently of those conditions.

### *4.1 Hermann-von Weizsäcker Microscope*

The starting point of Hermann's reflection is that the experimental impetus for the considerations that culminate in the assertion of insurmountable limits to predictability was provided by the so-called dualism experiments. These experiments showed that the classical distinction between processes consisting in the rapid motion of small material particles and processes involving wave propagation can no longer be maintained. Both electrons and photons must be described under the dual aspect of corpuscle and wave: "In this relative character of the quantum mechanical mode of description lies the new and amazing natural-philosophical aspect that is here introduced into the view of nature. What it consists of can best be understood on the basis of a thought experiment treated by a pupil of Heisenberg" (Hermann 1935a, 110f./257). The student in question is of course von Weizsäcker. In 1931, he published a paper on the localization of an electron by means of a microscope, developing a modified version of Heisenberg's famous microscope argument (Weizsäcker 1931). In Heisenberg's original version, the measurement of position leaves the electron's momentum indeterminate because of the uncontrollable momentum transfer involved in the scattering process. Von Weizsäcker's treatment, by contrast, emphasizes a different point: measurements of position and momentum require two incompatible experimental arrangements. Von Weizsäcker's aim was primarily to substantiate Heisenberg's analysis through detailed calculations; Hermann, by contrast, used a qualitative version of the thought experiment to express the central conceptual novelty of quantum mechanics (Filk 2017).

Since Hermann's presentation seems to presuppose some working knowledge of the apparatus, I will provide a more didactic account. Hermann considers an electron moving along the object plane $L$, identified with the $x$-axis. The relevant conjugate quantities are therefore its position $x$ along this plane and the corresponding momentum component $p_x$. To measure the electron, one illuminates it with a light quantum. The situation can be simplified by assuming that only a single photon in each run is involved. The central point is that the light quantum involved in the electron's illumination must be described under the dual aspect of wave and particle. As a particle, it collides elastically with the electron according to the classical laws of impact, where momentum conservation applies: both the light quantum and the electron are deflected, and their changes of momentum are equal in magnitude and opposite, in particular in the $x$-direction. As a wave, it is deflected by the electron, enters the microscope, is propagated through the lens according to the classical laws of optics, and reaches a scintillating screen $S$.

It is useful to spell out a few details from classical optics that remain implicit in Hermann's account. A lens takes parallel rays and bends them toward a single point. Let us assume that the lens has fixed optical properties, determined by its refractive index $n$ and by the radii of curvature $R_1$ and $R_2$ of its surfaces. These properties determine the focal length $f$, the distance from the lens to the focal point at which rays entering parallel to the optical axis are brought together. For a thin lens in air, this relation is given by the lens maker's equation:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$

The focal length $f$ is therefore a constant determined by the lens and the surrounding medium. It should be distinguished from the image distance, that is, the distance from the lens's principal plane to the image plane at which rays diverging from a given object

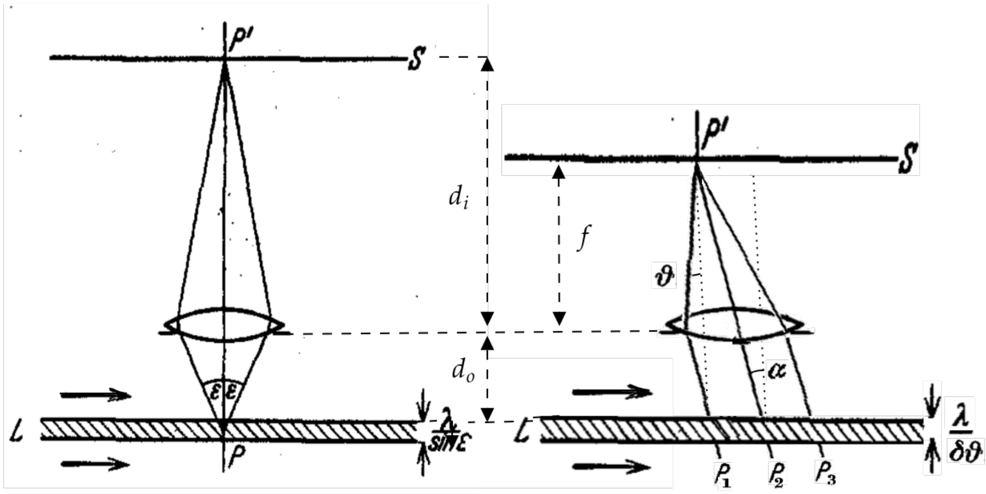


Figure 5: A schematic of the von Weizsäcker-Hermann microscope. Left (original from Weizsäcker 1931): $S$ in the image plane. Right (reconstructed): $S$ in the focal plane

point are brought together at the corresponding image point. The image distance depends not only on the lens but also on the position of the object plane $L$. If $d_o$ denotes the distance from $L$ to the lens, and $d_i$ the distance from the lens to the plane in which the corresponding image is formed, the thin-lens equation gives

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}.$$

Thus, once the lens and the object plane $L$ are fixed, $d_o$ is constant, and the corresponding image distance $d_i$ is thereby determined. The same lens can nevertheless be used in two distinct arrangements, depending on the position of the screen $S$ relative to the lens: (a) the lens-to-screen distance is $d_i$, so that $S$ lies in the *image plane* (b) the lens-to-screen distance is $f$, so that $S$ lies in the *focal plane* .

All the light entering the lens passes through both the focal and the image planes, but different families of rays come to coincidence in each at $P'$. In the first arrangement, points $P$ in the object plane $L$ are mapped onto corresponding points $P'$ in the image plane; in the second, rays arriving at the lens in the same direction are brought together at the same point $P'$ on the screen. For a real object beyond the focal length, the focal plane at $f$ precedes the image plane at $d_i$. The lens brings parallel rays to a focus at the focal distance. Rays spreading from a finite-distance object at $P$, however, arrive at the lens already diverging; the lens must first compensate for this divergence, so that they coincide only farther away, at the image distance $d_i$, after which they diverge again. Moving $S$ from $d_i$ to $f$ therefore means intercepting the refracted light field at a different stage of its propagation along the optical axis. This classical optical alternative constrains the operation of the von Weizsäcker-Hermann microscope (Hermann 1935a, §10), which functions as a selection or preparation device for incoming photons, either according to position or according to direction:

– $S -$lens $= d_i$: In the image-plane arrangement, the optical system groups together photons scattered from the same position $P$, even though they leave $P$ in different directions within the aperture. These directions are recombined at the image point $P'$. From the mark on the plate at $P'$, one can infer the position $P$ of the electron at the time of its collision with the photon. This description presupposes that a spherical wave spreads out from the point of collision and that the whole aperture of the microscope contributes to the image. In the corresponding corpuscular description, however, this means that no definite direction can be assigned to the photon after scattering. The momentum transferred to the electron therefore cannot be exactly determined. Immediately after the collision, the electron must be described by a wave function $\psi_E$ that assigns it a relatively sharp position but a less sharply determined momentum.

  - To obtain a very precise position measurement $\Delta x$, one needs a large lens aperture $2\epsilon$, so that the microscope collects more of the spherical wave coming from $P$ and thereby localizes its origin more precisely.[7] A large $\epsilon$, however, means that the scattered photon may have entered the microscope from within a wide range of directions. This produces a large, unavoidable uncertainty in the momentum transferred to the electron, so that $\Delta p_x$ is correspondingly large. In this arrangement, therefore, only the probability of finding the particle with a given direction of motion can be specified.

$$\Delta x \sim \frac{\lambda}{\sin\epsilon}, \quad \Delta p_x \sim \frac{h}{\lambda}\sin\epsilon.$$

- $S - \text{lens} = f$: In the focal-plane arrangement, the optical system groups together photons with the same outgoing direction, even though they may have been scattered from different points $P_1, P_2, P_3, \ldots$ in the object plane $L$. These photons are focused onto the same point of the focal plane. The point $P'$ at which the photon blackens the photographic plate $S$ therefore determines the direction in which the scattered light entered the microscope. The scattered photon travels mostly upward, so its momentum has a large $y$-component and, if it is slightly inclined, a smaller $x$-component. If $\alpha$ is the angle between the scattered photon's direction before it reaches the lens and the vertical optical $y$-axis, then the transverse momentum component of the photon is $p_{P,x} = p_P \sin\alpha$. If the photon's transverse momentum before the collision is known, conservation of momentum fixes the corresponding change in the electron's momentum $p_{E,x}$. In this arrangement, however, the point in the object plane from which the light originated, and hence the position at which the collision occurred, remains indeterminate. The electron wave function $\psi_E$ is therefore represented, in this momentum context, approximately as a plane wave with relatively sharp momentum but indeterminate position.

  - To obtain precise momentum information, the apparatus must distinguish closely spaced incident directions. The lens serves as the optical bridge, mapping the physical scattering angle $\alpha$ in the object space to the refracted angle $\vartheta$ in the image space. However, the resolving power of the focal-plane mark is limited to a narrow angular spread $\delta\vartheta$ of the refracted light.

[7] If the aperture angle $2\epsilon$ is small, the lens receives only a narrow cone of the spherical wave. Then many nearby points would produce nearly indistinguishable wavefronts at the lens, and the resulting image point is diffraction-broadened. The origin of the spherical wave is therefore only poorly localized.

This instrumental limit $\delta\vartheta$ determines the irreducible uncertainty $\delta\alpha$ of the initial scattering angle. The smaller $\delta\vartheta$ is, the more precisely the photon's original direction, and hence its transverse momentum component, can be known. This epistemic gain, however, leaves the point of origin in the object plane, and thus the position of the scattering event, indeterminate: only the probability of finding the particle at a given point can be specified.

$$\Delta p_x \sim \frac{h}{\lambda}\delta\vartheta, \quad \Delta x \sim \frac{\lambda}{\delta\vartheta}.$$

It is worth noting that Hermann's analysis presupposes two aspects of wave-particle duality that are not always clearly distinguished in her text. The first may be called *inter-contextual*: different optical (image or focal plane) arrangements correlate the registration with different physical magnitudes, either the point of origin or the direction of propagation, and hence either position or momentum. The second is *intra-contextual*, internal to each context: once an arrangement has been chosen (image or focal plane), the wave and corpuscular descriptions mutually restrict one another, yielding the corresponding uncertainty trade-off between position and momentum. (a) Bohr's *complementarity*, understood as an either-or choice between mutually exclusive contexts, is the expression of the inter-contextual aspect of the wave-particle duality; (b) Heisenberg's *uncertainty*, understood as expressing a gradual trade-off internal to each experimental context, is the expression of its intra-contextual aspect.

If no photographic plate $S$ is introduced, the photon is not registered, and a third description is required, which might be called *pre-contextual*, that is, prior to the later registration context (135/258). The incompatibility of the focal-plane and image-plane arrangements is not by itself sufficient to exclude a context-independent state, since the same operational incompatibility occurs in classical optics. The specifically quantum-mechanical premise is that, after the collision, the electron and photon are generally described by a non-factorizable joint wave function. Hermann explains this in a short note set in smaller type, worth examining in some detail:

– Before the collision, the two systems may still be described independently by a product state. After the collision, however, the interaction generally destroys this factorability. The post-collision state is not, in general, the product of a wave function for the electron $\psi_E$ and a wave function for the photon $\psi_P$. Rather, it is a linear combination, that is, a weighted sum, of product terms, each containing a wave function for the electron and one for the photon. Mimicking the notation of the period, one may write:

$$\psi_{EP}(x_E, x_P) = \sum_{mn} c_{mn}\psi^E_m(x_E)\psi^P_n(x_P).$$

This expression is factorizable only in the special case in which the coefficients can be written as $c_{mn} = a_m b_n$. In general, this condition is not satisfied. The collision has therefore produced a joint state in which the two systems are no longer characterized independently. In the idealized case of perfect one-to-one correlations, the state may be written in the form

$$\psi_{EP}(x_E, x_P) = \sum_{n} c_n\psi^E_{m(n)}(x_E)\psi^P_n(x_P).$$

The coefficients then determine the probabilities of the correlated pairs: $|c_n|^2$ is the probability that the photon system is found in the state $\psi_n^P$ and the electron system in the uniquely correlated state $\psi_{m(n)}^E$. Accordingly, if the photon is found in $\psi_n^P$, the electron is found with certainty in $\psi_{m(n)}^E$, so that $P(m(n) \mid n) = 1$.

The same joint wave function can be represented in different eigenfunction bases. If the registration plane is placed at the focal plane, one represents it in eigenfunctions of the relevant momentum operator. One then obtains a representation in which each term correlates a photon momentum eigenstate $\psi_p^P$ with the corresponding electron momentum eigenstate $\psi_{P_{\mathrm{tot}}-p}^E$, by momentum conservation:

$$p_E + p_P = P_{\mathrm{tot}}.$$

If, instead, the registration plane is placed at the image plane, one represents the same wave function in eigenfunctions of the position operator. One then obtains a representation in which each term correlates a photon position eigenstate with the corresponding electron position eigenstate, according to the geometrical relation established by the optical apparatus between the collision point and the screen mark:

$$x_E - x_P = \text{constant}.$$

The important point is that these are not two independent descriptions of two independently characterized systems. They are two representations of one and the same non-factorizable joint state. Depending on the experimental arrangement, the same state can be decomposed into terms correlating momentum eigenstates or position eigenstates.

The situation already exhibits the correlation structure that, as we shall see, would become central to the quantum-mechanical discussion within a few months: a joint state of two systems can be represented in different ways, each correlating a different pair of quantities. By *deciding* whether to measure the position or the momentum of the photon, one can predict with certainty either the position or the momentum of the electron. Hermann's point is that these two counterfactual descriptions cannot be combined into a real position-cum-momentum state of the electron, since they belong to different and incompatible contexts.

Hermann concludes that a registration event, a mark $P'$ on the photographic plate $S$, has no determinate physical meaning by itself. Its significance depends on the *reconstruction* of the optical arrangement (the lens, the focal point, the position of the planes , etc.) that correlates it either with a direction of propagation or with a point of origin. In particular, if the plate $S$ is placed in the focal plane, at distance $f$ from the lens, the lens maps parallel rays, and hence a definite propagation direction, onto a point on the plate. The mark $P'$ is then interpretable as momentum information. Its precision depends on the angular spread $\delta\vartheta$ of the rays contributing to that mark: the smaller $\delta\vartheta$, the more sharply the corresponding momentum component is determined. If, by contrast, the plate $S$ is placed in the image plane corresponding to the object plane, at distance $d_i$ determined by the thin-lens equation, the lens maps object points onto image points. The same kind of mark $P'$ is then interpretable as position information, since it indicates the point $P$ from which the photon was scattered. Its precision depends on the resolving power of the image, and thus on the aperture angle $2\epsilon$.

The measurement outcome is therefore not the mark $P'$ alone, but that mark together with the optical history connecting the scattering event to its registration on the plate. Here Hermann can articulate her position on causality after quantum mechanics. The principle of causality can no longer be understood to require sharp *predictions* of future measurement outcomes, since quantum mechanics generally yields only statistical predictions. Rather, it must be presupposed as the condition for unambiguous *retrodiction* from an already registered measurement outcome to the microscopic quantity. As in classical physics, this retrodiction requires a *causal reconstruction*: only through a theory of the measuring apparatus can one draw an unambiguous inference from a measuring result, such as a pointer reading, to a property of the object measured. However, quantum mechanics makes this dependence more radical. Because of wave-particle duality, in quantum mechanics one may reconstruct the measurement as yielding either a position determination or a momentum determination, but not both within a single observational context. The choice of context is therefore *free*, in accordance with wave-particle duality; once that context has been chosen, however, the reconstruction is *necessary*, as required by the principle of causality.

A few pages later—probably after her correspondence with von Weizsäcker about Popper—Hermann added a small-print note warning against confusing two types of causal reconstruction—legitimate 'measurement reconstructions' and illegitimate 'trajectory reconstructions':

> The retrospective inferences [*Rückschlüsse*] underlying these indirect predictions, however, must not be confused with those familiar inferences [*Rückschlüssen*] that apparently allow one to determine a physical system for a past time segment with accuracy exceeding the uncertainty relations. Thus it appears as though one could, for instance, escape the limits on measurement accuracy by taking two position measurements on an electron in quick succession, or by first determining its momentum and shortly thereafter its position, in order then to calculate from the results of both measurements together the exact trajectory of the electron for the time interval between the two measurements, and thus determine with arbitrary accuracy its respective position and momentum for this time. (Hermann 1935a, 121/262)

Hermann argues that the after-the-fact reconstruction of a particle trajectory differs fundamentally from the retrospective reconstruction of the measurement process. According to Hermann, *trajectory reconstructions* ignore the dualism of wave and particle descriptions; in particular, they ignore what I have labeled the inter-contextual aspect of this dualism:

> This posterior [*nachträgliche*] construction of an electron trajectory differs from the inferences [*Rückschlüsse*] that are necessary for the interpretation of a measurement in that it takes no account of the duality of the wave and particle picture, limits itself to the intuitive features of one of these pictures and, since each single observation makes reference to both pictures and hence admits unlimited use of neither, fixes these features through two different observations. In this way one apparently succeeds in combining the variables belonging to different observational contexts into a single description of the physical system. In contrast to this, the causal inference that determines the state of the observed from that of the measuring instrument must take duality into account in the interpretation of a measurement. For it is essential for the process of measurement that every atomic process also exhibits a wave character, and every wave motion also a corpuscular character. The single observation of the measuring instrument that grounds the inference to the object of measurement determines in which way the wave and particle picture delimit each other in the present interpretation. (Hermann 1935a, 121f./262)

The *trajectory reconstruction*, by combining features fixed through two different observational contexts, momentum *and* position, tries to unite determinations that in fact belong to different observational contexts within a single description of the physical system. The *measurement reconstructions*, from the mark of an indicator to the value of a quantity of the observed object, momentum *or* position, must account for wave–particle dualism. This, Hermann continues, was precisely the lesson of the microscope example:

> The significance of this contrast follows from the relative character of the quantum-mechanical mode of description. Since every physical description and explanation of processes is valid only relative to its respective observational context, the calculation of, say, a corpuscular trajectory that combines the variables of different observational contexts into a single representation remains physically vacuous precisely to the extent that it exceeds the uncertainty relations: it is uncheckable and provides no basis for future predictions.[*] The controversial representation Heisenberg gave of this situation—that it is purely a question of taste whether one should assign physical reality to such trajectory calculations—is therefore to be amended in the sense that such calculations disregard the quantum-mechanically crucial relation that each specification of a physical variable [*physikalische Bestimmung*] bears to the respective observational context and, in this sense, become physically meaningless. Conversely, the possibility of checking, even indirectly, the causal claims that arise in the interpretation of a measurement process depends on the fact that this interpretation does justice to duality, and thus implicitly to the relative character of quantum mechanics. (Hermann 1935a, 122/263)

The reference marked by the asterisk is again to p. 15 of Heisenberg's (1930) Chicago lectures. As we have seen, for Heisenberg, the question whether physical reality can be assigned to such a reconstructed trajectory is merely a 'matter of taste.' For Hermann, by contrast, it was, so to speak, a 'matter of context.' Like (Schlick 1931), Hermann argues that these reconstructions are *meaningless*. However, Hermann's contextualist reading is more subtle than Schlick's blunt verificationist approach. For Schlick, a past trajectory cannot be tested; for Hermann, the two counterfactual determinations cannot be combined. According to Hermann, at the inter-contextual level, the trajectory calculation ignores the quantum-mechanically decisive fact that every physical determination is relative to a specific observational context. Put in slightly modern terms,[8] one may construct either a $x(t)$-history or a $p(t)$-history, but not a hybrid history (*i.e.*, a trajectory) that treats both as simultaneously valid descriptions of the same individual process. At the intra-contextual level, once one of these histories has been selected, the alternative history can be reconstructed only probabilistically. If the chosen context fixes a position history, then the corresponding momentum history is not available as a determinate trajectory, but only as a distribution of possible momentum values. Conversely, if the chosen context fixes a momentum history, then the corresponding position history remains indeterminate and can be specified only probabilistically.

The premises for Hermann's critique of Popper are now established. She adds a footnote to the passage just quoted, marked here by an asterisk, applying this analysis to Popper's 1934 paper in *Die Naturwissenschaften*. As we have seen, Popper's strategy was based on a very permissive interpretation of Heisenberg's matter-of-taste remark: "One has attempted, however, to use such calculations of trajectories to derive predictions in a similarly indirect manner as happens in the interpretation and checking of measurements, and thereby to overcome the limits of the uncertainty relations" (Hermann 1935a, 123; fn. 10/263; fn. 10). As Hermann suggests, Popper's setup

[8] In the spirit of Griffiths (2003).

is structurally similar to Hermann's own. In both cases, after the electron–photon collision, the two particles are described by the same kind of pre-contextual joint state. Popper then *chooses* a context, that is, to measure first the momentum of the electron and later its position in order to reconstruct the trajectory and use conservation laws to predict the photon trajectory. Because Popper does not realize that this choice of context has been made, his attempt ultimately fails on two levels:

– At the intra-contextual level, after the electron–photon collision, a momentum measurement on the photon selects the momentum representation of the joint state. By momentum conservation, this also fixes the corresponding momentum of the electron. But precisely within this momentum context, the electron's position is indeterminate. Popper therefore cannot reconstruct the electron's position-cum-momentum trajectory *before* the first momentum measurement at $X$.

– At the deeper inter-contextual level, Popper tries to combine this momentum determination with a later position measurement, as if the two results belonged to a single classical history of the same particle. For Hermann, however, a momentum context and a position context cannot be combined in this way, but only used counterfactually. If one chooses the momentum context, one is cut off from the position context, and vice versa. Thus even the notion of a trajectory *between* the momentum measurement and the position measurement at $X$ has no well-defined quantum-mechanical meaning.

As already anticipated in her letter to von Weizsäcker, Hermann claims that the roots of Popper's mistake can be understood only if one reads not only Popper's 1934 paper but also his 1935 book: "the real basis of this mistake, as only the more detailed discussions in Popper's book *Logik der Forschung* (Vienna, 1935) show, lies in a misapprehension of the dualism experiments and their consequences" (263; fn. 10/123; fn. 10). Popper's statistical interpretation of quantum mechanics is the result of a misunderstanding of wave-particle duality at both the inter- and intra-contextual levels:

> Popper is led by the probabilistic interpretation of wave functions to apply these quantum-mechanical state descriptions and the uncertainty relations given with them, strictly speaking, only to ensembles of physical systems, while assuming that a suitably selected individual system is not bound by the uncertainty relations. In doing so, he fails to recognize that, on the basis of the dualism experiments, the applicability of classical concepts is already restricted for each individual elementary process in accordance with the uncertainty relations, and that, accordingly, wave functions are perfectly usable for describing the states of individual systems. That this use of wave functions is compatible with their probabilistic interpretation rests once again solely on the relative character of the quantum-mechanical mode of description: on the one hand, the wave function is completely determined by the values of those physical quantities that have a sharp value for the system in the present observational context. To this extent, it characterizes the system quantum-mechanically relative to the present observational context. On the other hand, the probabilistic interpretation of wave functions leads to those determinations that remain from the classical-intuitive description in accordance with the correspondence principle and that determine which statements can be made for the transition from one observational context to another. (123f.; fn. 10/263, f.)

The same point is reiterated in Hermann's review of Popper's book in *Physikalische Zeitschrift*, which appeared in July. As we have seen, the wave description imposes an inter-contextual restriction: one may not combine a sharp momentum value obtained in one context with a sharp position value that could only have been obtained in an incompatible context. Intra-contextually, the wave function quantifies the probability distribution for the variable that cannot be sharply determined within a given experimental arrangement. Popper's probabilistic interpretation is misleading because it remains *non-contextual*: it treats the wave function as a statistical description of an ensemble of *trajectories* that may still be conceived as possessing definite, context-independent joint position-and-momentum histories.

## 5 Conclusion: Hermann and Popper after the EPR thought experiment

Hermann's paper was completed in March 1935, the same month in which the EPR paper was submitted to the *Physical Review.* At roughly the same time, Popper came into contact with Einstein through a mutual acquaintance.[9] During the spring of 1935, Popper worked on a response to the criticisms from the Leipzig group, expanding it into a longer manuscript, again titled 'Zur Kritik der Ungenauigkeitsrelationen'. It cannot be ruled out that the appearance, in July, of Hermann's (1935b) review of the *Logik* in the *Physikalische Zeitschrift* reinforced his sense that it was necessary to take a public stance. Weisskopf's account of Einstein's recent critique of quantum mechanics may have led Popper to hope that Einstein would be a more receptive audience. He therefore decided to send him the manuscript (Popper to Einstein, Aug. 29, 1935; Albert Einstein Archives, 19–127).

Einstein replied in a letter of September 11. While acknowledging the general direction of Popper's argument, he nevertheless rejected the details of Popper's proposed experimental setup (Einstein to Popper, Sep. 11, 1935; Albert Einstein Archives, 19-130). Einstein reiterated to Popper what others had already told him: Whatever 'momentum filter' one uses, in the ideal limit, a procedure that selects a sharp momentum component would correspond to a state increasingly delocalized in the conjugate coordinate; hence it could not also provide the definite position information needed for Popper's trajectory reconstruction before the momentum measurement. At Popper's request, Einstein presented the argument that he had just published with two collaborators (Einstein, Podolsky, and Rosen 1935). The situation considered by Einstein is structurally close to the Hermann–Popper thought experiments: two systems, $A$ and $B$, interact for a short time and are then separated. After the interaction they are described by the joint non-factorizable wave function $\psi_{AB}$; a measurement is performed on $A$, and quantum mechanics assigns different $\psi_B$-functions to $B$ depending on which measurement has been chosen on $A$. In this sense, both Popper and Hermann came close to touching on what would become one of the central issues of the debate over the following decades.

However, Einstein's argument differs from both Hermann's and Popper's in a crucial respect. Einstein first presents Popper with a version of the argument that does not require the two systems to be perfectly correlated , or the $\psi_B$-function assigned to $B$ to be an eigenfunction of some operator (Einstein to Popper, Sep. 11, 1935; Albert Einstein Archives, 19-130). The state assigned to $B$ may still be a superposition. The only relevant fact is that different measurements on $A$ lead to different $\psi_B$-functions for $B$. Einstein then introduced a *criterion of sameness*: since a measurement on the now separated system $A$ cannot alter the real physical state of $B$, these different $\psi_B$-functions must correspond to one and the same physical state of $B$. The formalism of quantum mechanics, however, attributes two different wave functions to the same

[9]Busch to Einstein, Apr. 28, 1935; Albert Einstein Archives, 34-338; Einstein to Popper, Jun. 15, 1935; KPA 292-12; Popper to Einstein, Jul. 18, 1935; Albert Einstein Archives, 19-126.

physical state of $B$. Einstein concludes that the wave function cannot be a complete description of the physical state (Einstein to Popper, Sep. 11, 1935; Albert Einstein Archives, 19-130).

Einstein then describes the specifically EPR set-up, which appears as a special case of the preceding arrangement: the two systems are perfectly correlated, for example after a collision because of conservation laws (Einstein to Popper, Sep. 11, 1935; Albert Einstein Archives, 19-130). By choosing which measurement to perform on $A$, one can predict either the position or the momentum of the distant system $B$ with certainty, that is, with probability 1. If this prediction is made without disturbing $B$, then, according to the EPR 'criterion of reality,' the predicted quantity must already belong to the real physical state of $B$. Thus $B$ seems to possess both a definite position and a definite momentum, even though no single quantum-mechanical state function can represent both sharply. In this second formulation, Einstein's argument comes much closer to what Popper wanted: although Einstein cannot predict the future trajectory of the second particle, he seems to suggest that it nevertheless possesses, at the relevant time, both a definite position and a definite momentum.

Popper seems not to have realized the difference between these two arguments and quickly embraced the second EPR version as a 'weaker' replacement of his own thought experiment against the uncertainty relations. Hermann, of course, wrote before EPR. Yet her thought experiment is, after all, a kind of delayed-choice version of the latter (Jammer 1974). Bohr's (1935) response to the EPR paper, perhaps not surprisingly, seems to share the same argumentative structure (Bacciagaluppi 2017): since the experimental arrangements for measuring position and momentum are mutually exclusive, the counterfactual predictions concerning system $B$ are not two determinations made within a single experimental context. The possibility of predicting a quantity with certainty, without mechanically disturbing system $B$, therefore does not entail that this quantity belongs to $B$ as a context-independent element of physical reality. Heisenberg's insistence on having an *Auszug* of Hermann's article published in *Die Naturwissenschaften* by the end of the year suggests that it may have been intended as an intervention in the EPR debate (Bacciagaluppi and Crull 2024, 43).

Hermann's role, however, went unappreciated. The reception of Bohr's (1935) response to EPR among German-speaking philosophers did not proceed along neo-Kantian but positivist lines (Blumberg and Feigl 1931). Bohr's invitation to the Second International Congress for the Unity of Science held in Copenhagen in June 1936 was supposed to seal the alliance with the Vienna Circle (Faye 2010). Frank's (1936) and Schlick's (1936) found in Bohr's 'operationalist' appeal to the mutual exclusivity of experimental arrangements a means of reinforcing the unity of the otherwise somewhat fragmented Viennese response to quantum mechanics under the banner of 'verificationism' (Beller and Fine 1994). Popper was also invited to the Congress at the last minute and had the opportunity to discuss with Bohr thanks to Weisskopf's mediation (Weisskopf to Popper, Oct. 17, 1936; Karl Popper Archives 360-21). At the Congress, Hermann (1936) delivered a short address in response to Schlick, who had recently died (Bacciagaluppi 2020). Popper and Hermann may have had another opportunity to meet there, but there is no evidence that the encounter left any trace. Hermann and Popper both left continental Europe shortly thereafter, in 1937. They returned to quantum mechanics only around 1950: Hermann with a short paper Henry-Hermann 1948, and Popper with a series of talks on indeterminism in England and the United States between 1948 and 1950 (Popper 1950). Bohr and Einstein apparently attended the Princeton talk (Jammer 1991). Whereas Hermann seems to have moved on, absorbed in political and pedagogical commitments, for Popper this marked the beginning of a full-scale return to quantum mechanics after the debacle of the 1930s (Del Santo 2019).

Popper began writing the material on quantum mechanics for the Postscript to *Logik der Forschung* in 1951–56, shortly before the publication of the first English edition of *Logik* in 1959. Part of this material was included in the appendices, while the rest became a separate *Postscript*, which reached galley proofs in 1956–57 but remained unpublished. In the new English edition of *Logik*, the chapter on quantum mechanics was left untouched, but Popper sought to make amends for his previous mistakes with new footnotes and appendices marked with starred numbers. Popper acknowledged what von Weizsäcker had pointed out to him: trajectory reconstruction was not possible "*before* the first of these measurements", that is, the momentum measurement. Without this, Popper's thought experiment fails. However, Popper continued to hold that "non-predictive measurements determine the path of a particle only *between* two measurements," such as a measurement of momentum followed by one of position, or vice versa. (Popper 1959, 242; fn. *3; my emphasis). Popper seems not to have appreciated, and quite possibly not to have read, Hermann's deeper critique that trajectory reconstructions were impossible even in general. This is not simply an oversight.

Popper appears never to have abandoned the conviction that the very notion of past trajectories was not merely a 'matter of taste,' nor was it meaningless, as Bohr and the Copenhagen interpretation maintained. They are an essential component of quantum mechanics. In the main body of the *Postscript*, and more clearly in a 1966 text later included in expanded form in the printed volume, Popper argued that in quantum mechanics *predictions* of future trajectories are statistical, with dispersions related by the Heisenberg formulae; however, to test these predictions, one needs measurements, and "these measurements are retrodictions" (Popper 1967, 25; 1982, 60); such retrodictions are trajectory reconstructions. Thus quantum theory cannot exclude the possibility that an electron can 'have' a precise position and momentum in the past. This, according to Popper, is what EPR shows: the EPR paradox "is not a paradox but a valid argument, for it established just this: that we must ascribe to particles a precise position and momentum, which was denied by Bohr and his school" (63). Indeed, Popper could claim that he was among the first philosophers to realize the significance of EPR (14). Even without knowing whether Popper took notice of Hermann's critique of the very notion of trajectory in quantum mechanics, we can surmise how he would have reacted by looking at his dismissal of Bohr's reply to EPR.

For Popper's (1982, III-17), the EPR argument shows convincingly that one can move from two counterfactual attributions of position and momentum to one simultaneous state attribution. If this is possible, then it is natural for Popper to conclude that trajectory attribution should also be possible, since a trajectory is the ordered sequence of such state-attributions over time. Popper found Bohr's (1935, 1948) argument against EPR obscure, but he saw its thrust clearly: Bohr denies precisely that the two counterfactual attributions of position and momentum can be combined, since they belong to mutually exclusive experimental frameworks: "The arrangements are exclusive of each other (or 'complementary'), the argument continues, and so we cannot combine the two spaces [...] into one" (Popper 1982, 149). If simultaneous state attribution is denied, a continuous, context-independent history of position and momentum is also 'meaningless.' This was precisely the argument Hermann raised against Heisenberg's matter-of-taste remark. The Popper-Hermann debate ultimately revolved around this point: to test the theory we need measurements, and 'measurements are retrodictions,' as Popper put it. As Hermann objected, however, these retrodictions in quantum mechanics are not context-independent 'trajectory reconstructions,' but context-dependent 'measurement reconstructions.'

## References


Bacciagaluppi, Guido. 2017. "Bohr's Slit and Hermann's Microscope." In Crull and Bacciagaluppi 2017, 135–147.

Bacciagaluppi, Guido. 2020. "Translation of Three Short Papers by Grete Hermann." *Journal for General Philosophy of Science* 51:615–619.

———. 2025. "Better than Bohr: Grete Hermann and the Copenhagen Interpretation." Talk, slides in the author's possession, Torino, 2025.

Bacciagaluppi, Guido, and Elise Crull. 2024. *The Einstein Paradox: The Debate on Nonlocality and Incompleteness in 1935.* Cambridge, UK: Cambridge University Press.

Beller, Mara, and Arthur Fine. 1994. "Bohr's Response to EPR." In *Niels Bohr and Contemporary Philosophy,* edited by Jan Faye and Henry J. Folse, 153:1–31. Dordrecht: Springer Netherlands.

Blumberg, Albert E., and Herbert Feigl. 1931. "Logical Positivism." *The Journal of Philosophy* 28:281–296.

Bohr, Niels. 1928. "The Quantum Postulate and the Recent Development of Atomic Theory." *Nature* 121:580–590.

———. 1935. "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" *Physical Review* 48:696–702.

———. 1948. "On the Notions of Causality and Complementarity." *Dialectica* 2:312–319.

Cassidy, David C. 2015. "Werner Heisenberg and Carl Friedrich Freiherr von Weizsäcker: A Fifty-Year Friendship." *Physics in Perspective* 17:33–54.

Crull, Elise. 2022. "Grete Hermann's Interpretation of Quantum Mechanics." In *The Oxford Handbook of the History of Quantum Interpretations,* edited by Olival Freire et al., 567–586. Oxford: Oxford University Press.

Crull, Elise, and Guido Bacciagaluppi, eds. 2017. *Grete Hermann: Between Physics and Philosophy.* Berlin: Springer.

Del Santo, Flavio. 2019. "Karl Popper's Forgotten Role in the Quantum Debate: At the Edge between Philosophy and Physics in 1950s and 1960s." *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics* 67:78–88.

Einstein, Albert. 1932. "Über die Unbestimmtheitsrelationen." *Zeitschrift für angewandte Chemie* 45:23. Anonymous report of the Berlin colloquium on November 4, 1931.

Einstein, Albert, et al. 1931. "Knowledge of Past and Future in Quantum Mechanics." *Physical Review* 37:780–781.

Einstein, Albert, et al. 1935. "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" *Physical Review* 47:777–780.

Faye, Jan. 2010. "Niels Bohr and the Vienna Circle." In *The Vienna Circle in the Nordic Countries,* edited by Jutta Manninen and Friedrich Stadler, 14:33–45. Dordrecht: Springer.

Filk, Thomas. 2017. "Carl Friedrich von Weizsäcker's 'Ortsbestimmung eines Elektrons' and Its Influence on Grete Hermann." In Crull and Bacciagaluppi 2017, 71–83.

Frank, Philipp. 1932. *Das Kausalgesetz und seine Grenzen.* Wien: Springer.

———. 1936. "Philosophische Deutungen und Mißdeutungen der Quantentheorie." *Erkenntnis* 6:303–316.

Frappier, Mélanie. 2017. "In the No-Man's-Land Between Physics and Logic: On the Dialectical Role of the Microscope Experiment." In Crull and Bacciagaluppi 2017, 85–105.

Giovanelli, Marco. 2026. "The Past of an Electron: Young Popper's *Gedeankenexperiment* Against the Indeterminacy Relations." *Physics in Perspective.*

Griffiths, Robert B. 2003. *Consistent Quantum Theory.* Cambridge: Cambridge University Press.

Heisenberg, Werner. 1927. "Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik." *Zeitschrift für Physik* 43:172–198.

———. 1930. *Die physikalischen Prinzipien der Quantentheorie.* Leipzig: S. Hirzel.
Henry-Hermann, Grete. 1948. "Die Kausalität in der Physik." *Studium Generale* 1:375–383.
Hermann, Grete. 1933. "Determinismus und Quantenmechanik." Manuscript, Churchill/DRAC 3/11, The Papers of Professor Paul Dirac, Churchill Archives Centre, Churchill College, Cambridge. Published in Herrmann (2019, 185–203).
———. 1935a. "Die naturphilosophischen Grundlagen der Quantenmechanik." *Abhandlungen der Fries'schen Schule* 6:69–152. Tr. in Crull and Bacciagaluppi 2017, 239–278.
———. 1935b. "Review of Popper, *Logik der Forschung.*" *Physikalische Zeitschrift* 36:481–482.
———. 1936. "Zum Vortrag Schlicks." *Erkenntnis* 6:342–343.
Herrmann, Kay, ed. 2019. *Grete Henry-Hermann: Philosophie – Mathematik – Quantenmechanik: Texte zur Naturphilosophie und Erkenntnistheorie, mathematisch-physikalische Beiträge sowie ausgewählte Korrespondenz aus den Jahren 1925 bis 1982.* Wiesbaden: Springer.
Howard, Don. 2012. "Popper and Bohr on Realism in Quantum Mechanics." *Quanta* 1:33–57.
Jammer, Max. 1974. *The Philosophy of Quantum Mechanics: The Interpretations of Quantum Mechanics in Historical Perspective.* New York: John Wiley & Sons.
———. 1991. "Sir Karl Popper and His Philosophy of Physics." *Foundations of Physics* 21:1357–1368.
Kalckar, Jørgen, ed. 1985. *Niels Bohr's Collected Works: Foundations of Quantum Physics I (1926-1932).* Vol. 6. Amsterdam: North-Holland.
Mehra, Jagdish, and Helmut Rechenberg. 2001. *The Historical Development of Quantum Theory: The Completion of Quantum Mechanics, 1926–1941.* Vol. 6. New York: Springer.
Milkov, Nikolay. 2012. "Karl Popper's Debt to Leonard Nelson." *Grazer Philosophische Studien* 86:137–156.
Popper, Karl R. 1934a. "Zur Kritik der Ungenauigkeitsrelationen." *Naturwissenschaften* 22:807–808. With a reply by Carl Friedrich von Weizsäcker.
———. 1934b. "Zur Kritik der Ungenauigkeitsrelationen, 2. Mitteilung [A]." First pub. in Popper 2006, Doc. 10.
———. 1934c. "Zur Kritik der Ungenauigkeitsrelationen, 2. Mitteilung [B]." First pub. in Popper 2006, Doc. 11.
———. 1935. *Logik der Forschung: Zur Erkenntnistheorie der modernen Naturwissenschaft.* Wien: Springer. Tr. (Popper 1959).
———. 1950. "Indeterminism in Quantum Physics and in Classical Physics." *The British Journal for the Philosophy of Science* 1:117–133.
———. 1959. *The Logic of Scientific Discovery.* London: Hutchinson & Co.
———. 1967. "Quantum Mechanics without "The Observer"." In *Quantum Theory and Reality,* edited by Mario Bunge, 7–44. Berlin.
———. 1982. *Quantum Theory and the Schism in Physics: From the Postscript to the Logic of Scientific Discovery.* Edited by W. W. Bartley. London: Routledge.
———. 2006. *Frühe Schriften.* 1st ed. Edited by Troels Eggers Hansen. Vol. 1. Tübingen: Mohr Siebeck.
Reichenberger, Andrea. 2026. "Grete Hermann's Transformation of Neo-Kantianism: From Quantum Physics to Ethics." *HOPOS: The Journal of the International Society for the History of Philosophy of Science* 16:81–107.
Rosenfeld, Léon. 1967. "Niels Bohr in the Thirties." In *Niels Bohr: His Life and Work as Seen by His Friends and Colleagues,* edited by Stefan Rozental, 114–137. Amsterdam: North-Holland.
Schlick, Moritz. 1931. "Die Kausalität in der gegenwärtigen Physik." *Die Naturwissenschaften* 19:145–162.
———. 1936. "Quantentheorie und Erkennbarkeit der Natur." *Erkenntnis* 6:317–326.
Soler, Léna. 2017. "The Convergence of Transcendental Philosophy and Quantum Physics: Grete Henry-Hermann's 1935 Pioneering Proposal." In Crull and Bacciagaluppi 2017, 55–69.
Weizsäcker, Carl Friedrich von. 1931. "Ortsbestimmung eines Elektrons durch ein Mikroskop." *Zeitschrift für Physik* 70:114–130.